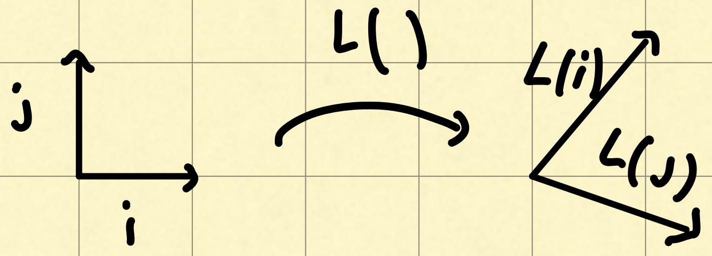
The Determinant

- lin transformations represented as matrices ✓
- Some stretch, some squish
- how much are things stretched or squished?
- Area → $L(\vec{v})$ → c · Area
  ↑ factor
- if you how much 1 is changed, you them all
- the scaling factor by which a linear transformation squishes space = the determinant of that transformation

- $\det\left(\begin{smallmatrix}\text{Lin}\\ \text{tra.}\end{smallmatrix}\right) = 3$ if lin trans stretches area by 3
- $\det(0) =$ if squished all of space onto a line/pnt
- check if $\det(0) =$ does $L(\vec{v})$ squish everything into smaller dimension?

- neg determinant: orientation flipped
- any transformation that swap the axes, "invert the orientation of space"



orientation changed, neg determinant

- but $|\det(\ldots)|$ = scale factor

- 3D determinant = how much volume scaled

- if cols of matrix are lin dep, det(0) ⇒ volume shrunk to plane

- $\det(2\times 2) = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$

- $\det(3\times 3) = \det\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = a \cdot \det\begin{pmatrix} e & f \\ h & i \end{pmatrix} - b \cdot \det\begin{pmatrix} d & f \\ g & i \end{pmatrix}$

$+ c \cdot \det\begin{bmatrix} d & e \\ g & h \end{bmatrix}$

. Explain in one sentence

$$\det(M_1 M_2) = \det(M_1) \cdot \det(M_2)$$

- bc the |det| is the scale factor by which a lin tran scales area (in R2), matrix multiplication is a composition of multiple transformations. ∴ the product of determinants = det of product.

- but at the same time, $M_1 M_2 \neq M_2 M_1$ ...
  → matrices produced are not the same, but the area of the matrices is.

→ diff transforms, same area

- matrix multiplication is function composition